



Electrical and mechanical properties of poly(ethylene oxide)/intercalated clay polymer electrolyte

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ABSTRACT

Solvent-free solid polymer electrolytes (SPEs) based on two different poly(ethylene oxide), PEO Mw 600,000 and 4,000,000 and intercalated clays are reported. The inorganic additives used were lithiated bentonite and the nanocomposite PEO@bentonite with the same polymer used as matrix. SPE films, obtained in the scale of grams by mixing the components in a Brabender-type batch mixer and molding at 130 °C, were characterized by X-ray diffraction analysis, UV–vis spectroscopy, and thermal analysis. During the preparation of the films, the unmodified clay got intercalated *in situ*. Comparative analysis of ionic conductivity and mechanical properties of the films show that the conductivity increases with the inclusion of fillers, especially for the polymer with low molecular weight. This effect is more pronounced when using PEO@bentonite as additive. Under selected work conditions, avoiding the presence of crystalline lithium complexes, observed effects are mainly centered on the polymer. An explanation, considering the higher affinity between the modified clay and PEO matrix which leads to differences in the micro homogeneity degree between both types of polymer electrolytes is proposed.

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1. Introduction

During the last decades the interest on solid polymer electrolytes (SPEs) has grown, together with the search of new, more efficient, high performance devices for energy conversion and storage, demanded by increasing need of producing clean energy [1]. Electrical conductivity and the mechanical and chemical stability are, in general, the most important properties of materials to be used as electrolytes in solid-state devices. That notwithstanding and depending on the type of applications, other properties, such as flexibility or transparency to a given light wavelength, may also be important. Attaining products with controlled properties requires however deep knowledge on the variables and of the mechanisms involved in studied system as well as on the conditions at which they are effective.

Polyethers, particularly poly(ethylene oxide) (PEO), have played a major role in the development of polymer electrolytes for lithium

batteries. The affinity of ether oxygen for lithium ions leads to polymeric coordination complexes able to function as lithium conducting solids [2,3]. Much theoretical and experimental work has helped in understanding ion conduction in these highly disordered systems [4–8]. Conductivity in these polymeric systems is essentially associated with ion migration between coordination sites, but also with segmental motions of the polymer chains. Thus, the most important variables determining performance of SPEs are lithium ion activity and the degree of amorphousness of the polymer. Unfortunately PEO-based materials often display poor conductivity at room temperature because of reduced segmental and chain mobility due to the semicrystalline nature of the polymer. So the development of these systems has been mainly marked by different attempts to increase ion mobility. This, however, must be accomplished without affecting the mechanical properties and dimensional stability of the material as may occur, for instance, with the use of low molecular weight plasticizers or high temperature work conditions. Although interactions with the lithium salt occur at the expense of polymer–polymer interactions decreasing its crystallinity, an excess of lithium ions could lead to the formation of new crystalline complexes with higher Li/polymer stoichiometric ratios, thus decreasing lithium ion activity. Therefore, SPEs based on poly(ethylene oxide) are complex systems in

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which there is an equilibrium of phases which depends, among others, from both lithium concentration and temperature. In the case of the semicrystalline system PEO/LiClO₄ at room temperature there are crystalline domains in which pure PEO can coexist with the crystalline complexes (PEO)₆LiClO₄ and (PEO)₃LiClO₄ [9] up to a molar ratio ethylene oxide (EO) to lithium ion (EO:Li⁺) of about 6:1, while at lower concentration of lithium only the crystalline complex (PEO)₆LiClO₄ could be present in the solid phase. However, the latter also depends on the thermal history of the sample. Recently [10] it was indeed demonstrated that the crystalline phase of this complex may form at room temperature by cooling the melt, but not immediately, being required an annealing period. So it is expected that when the solid polymer electrolyte film is prepared by diffusion of the lithium salt from a solution at room temperature, any contribution to the crystallinity of the sample of eventual PEO/Li⁺ crystalline complexes would be not significantly important.

An interesting attempt to reduce the crystallinity and enhance the conductivity of polymer electrolytes has been recently proposed, namely the design of composite polymer electrolytes (CPEs), by introducing inorganic fillers, for instance Al₂O₃ or TiO₂ nanoparticles [2,11] in the polymer matrix, where they play the role of “solid plasticizers” [12]. The presence of these ceramic elements under particular conditions, generally within narrow concentration ranges, may optimize the Lewis acid–base interactions between the species involved in order to reduce polymer–polymer interactions and, simultaneously, to increase the activity of free lithium ions favoring their mobility. However, the mechanism by which this kind of fillers enhances conductivity, particularly in complex systems like PEO-based SPEs, remains still unclear. Studies of systems like that reported recently by Fullerton-Shirey and Maranas [13] concerning the PEO/Al₂O₃ CPE, performed under conditions by which it is possible to identify the effect of the additive, separately on the different crystalline phases, are especially interesting. Among numerous materials tested as fillers, the use of layered clays, for instance montmorillonite or hectorite, has also been reported [14,15]. This type of clays appears to be particularly promising due to its cationic exchange capacity as well as to its ability for participating in intercalation and swelling processes. The latter certainly facilitates the dispersion of the filler within the polymer matrix, but it may be also used for improving and/or regulate the properties of the electrolyte by means of an adequate functionalization. Indeed it has been reported [16] that the activity of organo-clays as fillers increases with increasing inter-laminar distances in the intercalated clay. Furthermore, we have recently demonstrated that the use of lamellar fillers intercalated with the same polymer electrolyte of the matrix enhances lithium conductivity. Thus, in polyacrylonitrile-based polymer electrolytes using the filler bentonite-Li⁺–polyacrylonitrile, ionic conductivities about one order of magnitude higher than those with bentonite-Li⁺ under the same conditions were attained [17]. A similar effect has been also observed in a PEO/LiClO₄ based solid electrolyte using an organic–inorganic hybrid as filler, namely mesoporous silica (SBA-14) functionalized with the poly(ethylene oxide)–poly(propylene oxide)–poly(ethylene oxide) triblock copolymer EO₂₀PO₇₀EO₂₀ (P125) [18].

In this work we describe the effect on the conductivity and mechanical properties at room temperature of PEO/3 wt% clay-based composite polymer electrolytes produced by active fillers constituted by bentonite-Li⁺ intercalated with the same polymer used as matrix in the electrolyte. The study specifically concerns a PEO-based solid polymer electrolyte prepared using selected conditions under which, regardless of the SPE conductivity, the crystallinity of the system mainly arises from the PEO phase and not from any PEO/Li⁺ crystalline complex. Results indicate that the addition of the hybrid nanocomposites PEO@bentonite-Li⁺

Table 1

Composition of the films (wt%).

Sample	Bentonite-Li ⁺	PEO@bentonite-Li ⁺	PEO Mw 600,000	PEO Mw 4,000,000
1	3	–	97	–
2	–	3	97	–
3	3	–	–	97
4	–	3	–	97

improves the optical, mechanical and electrochemical properties of the electrolyte in relation to those attained using bentonite-Li⁺ as filler.

2. Experimental

2.1. Purification and characterization of the clay

Bentonite (Sigma) was first purified by cation exchange and sedimentation techniques and then converted to its lithium form by ionic exchange with 1 M lithium chloride. Salt excess was separated by dialysis (Sigma dialysis tubing cellulose membrane). The mean particle size of the product after filtering the suspension with a membrane (Whatman glass micro fiber filters, GF/D) was less than 2 μm. The composition of the montmorillonite, bentonite-Li⁺, M_{0.47}(Al_{1.58}Fe_{0.17}Mg_{0.25})(Al_{0.22}Si_{3.78}O₁₀)(OH)₂·*n*H₂O (M:Li) was determined by plasma atomic absorption spectroscopy (Perkin Elmer ICP-OES OPTIMA 1000DV). Cation exchange capacity (*cec*), 96.6 meq per 100 g of the clay, was determined using an ethylenediamine complex of Cu(II); concentration of Cu was determined by iodometry.

2.2. Preparation of the intercalate bentonite-Li⁺–PEO

5 g bentonite-Li⁺ was dissolved in 500 ml of 0.1 M solution of PEO (Mw 600,000–4,000,000 Aldrich, 97% and 98% respectively) by stirring for 3 days. Then the mixture was sonicated for 8 h and the solid product was separated by centrifugation. The solid extracted bentonite-Li⁺–PEO was washed 6 times with distilled water to remove excess PEO and then dried in a vacuum line for 3 days at room temperature.

2.3. Preparation of SPE films

Both kinds of films, PEO and PEO/clay SPEs, were produced by direct melt mixing in a Brabender-type batch mixer, at 130 °C and 75 rpm, for 15 min and then, before solidifying, flattened in a cold hydraulic press to produce sheets of about ca. 5 mm thick. Each mixture, having a mass of approximately 30 g, was composed of 29.1 g poly(ethylene oxide) (Mw 600,000 or 4,000,000) and 0.9 g of filler (3 wt%). The composition of the films is reported in Table 1. Polymer and composite films used for mechanical tensile testing and conductivity measurements were prepared by melt pressing the samples at 25 °C above the melting temperature of the material at 50 bar for 3 min in a heated hydraulic press using a mold appropriate for controlling thickness. Films were always 0.2 ± 0.02 mm thick. The specimens were cooled while pressing by flushing the press with cold water.

2.4. Mechanical property testing

SPE films with dimensions of 30 mm × 5 mm × 0.2 mm were cut using a heated stainless steel die in a hydraulic press. Among at least 5 specimens cut from each composite sheet, the three most homogeneous samples were selected for tensile testing. Mechanical properties were measured using a HP D500 dynamometer at a

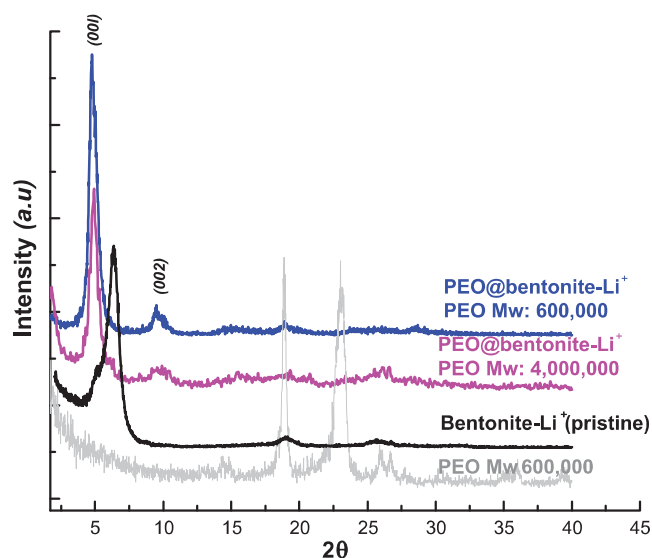


Fig. 1. X-ray diffraction patterns of films of PEO, bentonite-Li⁺ and nanocomposites PEO@bentonite with PEO Mw 600,000 and Mw 4,000,000.

rate of 50 mm/min at $27 \pm 1^\circ\text{C}$. Young's modulus was determined from the slope considering the three first points. The stress at break was defined as the last point prior to the precipitous drop in strain associated to sample breaking.

2.5. Electrical conductivity testing

The films were dipped into a LiPF₆–PPC 1 M solution for 24 h, to promote both dispersion and incorporation of the lithium salt, washed with methanol, and dried at 40°C under vacuum for 48 h. Average lithium content in the films corresponded to a ratio EO:Li⁺ of about 35:1.

The electrical conductivity of the films, sandwiched between ion-blocking gold electrodes, was measured by impedance electrochemical technique in an Autolab Potentiostat/Galvanostat-12 and FRA 2 equipment at 25°C .

X-ray powder diffraction data was collected on a Siemens D5000 powder diffractometer, using CuK radiation. FT-infrared spectra were obtained in a Bruker Vector 22 Spectrometer using 5% (w/w) pressed KBr disks. Transmittance of the membranes with thicknesses about 250 μm was measured with a UV–Vis spectrometer SHIMADZU 2450.

3. Results and discussion

3.1. Nanocomposite PEO–bentonite-Li⁺

The treatment of a suspension of the bentonite in water with a solution of PEO in the same solvent leads to a single lamellar phase. Elemental analyses indicate polymer intercalation ratios of 1.0 and 1.4 of monomer per clay structural unit, Si₄O₁₀, for PEO with molecular weight 600,000 and 4,000,000 respectively. The corresponding stoichiometries of both nanocomposites are therefore Bent_{1.0}Li_{0.4}(PEO)_{1.0}·1.2H₂O and Bent_{1.0}Li_{0.4}(PEO)_{1.4}·0.4H₂O respectively. Fig. 1 shows diffraction patterns for PEO (Mw 600,000), bentonite-Li⁺, as well as those of the nanocomposites bentonite-Li⁺–PEO with molecular weight 600,000 and 4,000,000 respectively. In both nanocomposites two 001 Bragg's reflections, corresponding to aluminosilicate interlayer separations of 18.6 Å, and 18.0 Å respectively may be observed. Diffraction patterns show the presence of a single laminar phase in both cases. Furthermore, the absence of diffraction lines at 2θ values between 15° and 30°,

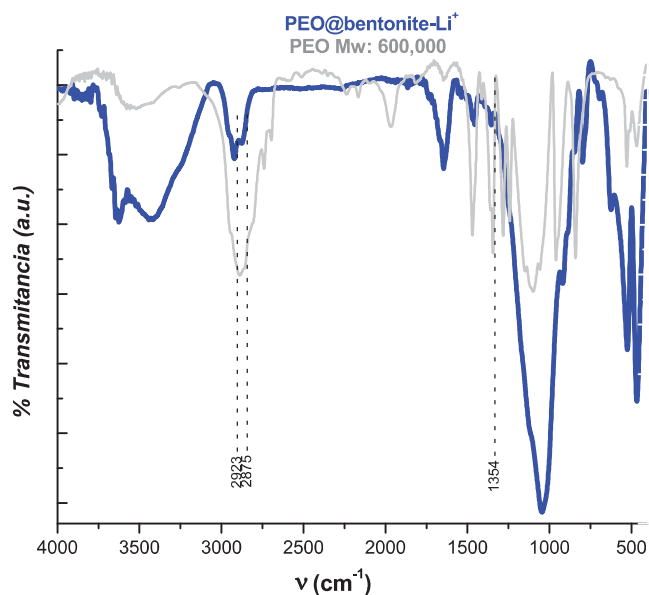


Fig. 2. FT-IR spectra of the polymers and the SPE with PEO@bentonite-Li⁺ with PEO Mw 600,000.

characteristics of poly(ethylene oxide), indicates that intercalated polymer is found in an amorphous phase. These products appear to be commensurate nanocomposites, so the difference of intercalated polymer between both nanocomposites is probably due to further displacement of water, together with a greater amount of dangling polymer chains at the edges of clay particles occurring in the case of the polymer with higher molecular weight. These results agree with earlier reports [19,20]. Infrared spectra reproduced in Fig. 2 show characteristic absorption bands of both clay and polymer, corroborating the organic–inorganic nature of the nanocomposites. A typical triplet peak of the C–O–C stretching vibrations with maxima at about 1149, 1110, and 1062 cm⁻¹ [21,22] is also clearly detected. Significant changes are observed in the wide absorption band associated to symmetric stretching of methyl groups in the range 2940–2840 cm⁻¹ which is observed in the nanocomposite as a well defined doublet centered at 2923 and 2875 cm⁻¹. This splitting, together with the shift of the torsion of methylene groups, from 1352 cm⁻¹ in free polymer to 1354 cm⁻¹ in the confined one, suggests a “gauche” conformation [23]. These spectra agree with the two types of conformations more probable for the polymer when is inserted in the interlayer space, i.e. in a zigzag planar conformation and a helical one. A study of the conformation of PEO intercalated into vanadium oxide, investigated by calculating 1-D electron density maps along the c-axis of the nanocomposite [24], arrives to the conclusion that the PEO chains are arranged side-by-side, in a fully extended conformation, across the interlamellar space.

Studies on self-assembling of poly(ethylene oxide) intercalated in molybdenum disulfide by ⁷Li and ¹H solid-state NMR [25] lead to similar conclusions. Indeed, results indicate that the conformation of PEO chains inside the MoS₂ interlayer space correspond to the sequence TGG/TGG/TGG, with torsion angles $T = 180^\circ$ and $G = \pm 75^\circ$. As stated earlier in the literature [24,26], the conformation of the polymer in this kind of systems appears to be associated with the magnitude of the interlamellar expansion induced by the intercalation. Therefore structural and compositional experimental data discussed above were analyzed considering a geometric criterion, assuming for the polymer a conformation similar to that found in the MoS₂ nanocomposites [25]. Results of this analysis are summarized in the schema illustrated in Fig. 3. Molecular models of PEO–Li⁺ complexes, built by using atomic van der Waals radii,

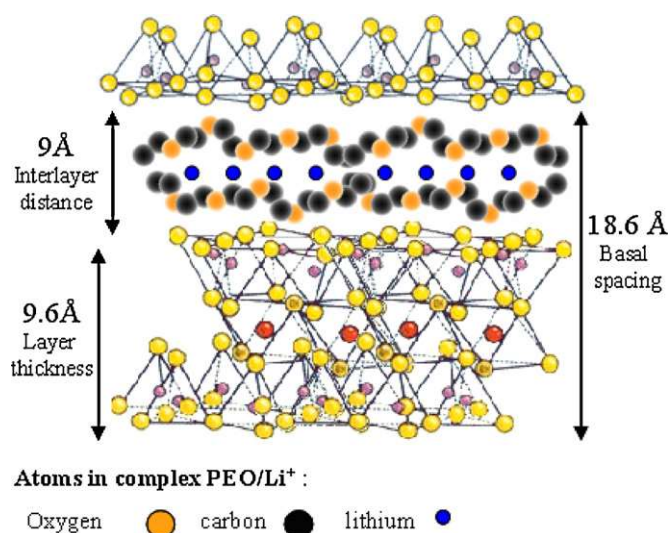


Fig. 3. Schematic description of the structure of the nanocomposites PEO@bentonite-Li⁺.

generate volumes which are in agreement with the interlaminar distances as well as with the surface available in the interlayer space of the clay.

3.2. Composite polymer electrolytes

Easily reproducible homogeneous films of PEO, pure and modified with clay fillers, were prepared by a batch process in a Brabender type batch mixer and molding at 130 °C as described above. Polymer electrolytes were then obtained by impregnation of the film by immersion in aqueous LiPF₆ solutions during 24 h at room temperature.

3.2.1. X-ray diffraction analysis

Diffraction patterns of membranes of PEO, PEO/bentonite-Li⁺ and PEO/PEO@bentonite-Li⁺ nanocomposite using PEO with molecular weight 600,000 are compared with that of the nanocomposite in Fig. 4. In the diffraction pattern of PEO/bentonite-Li⁺ SPE no basal peak of bare bentonite can be detected; instead, a diffraction peak at $2\theta = 5.16^\circ$, corresponding to a basal spacing of 17.1 Å is found. This fact clearly reveals that under our working conditions the

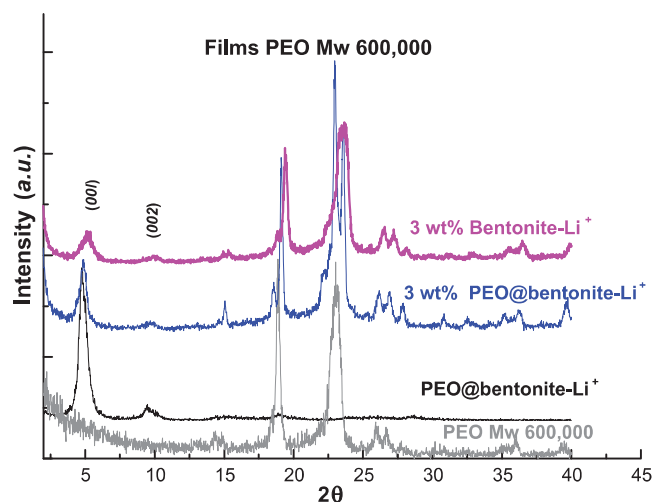


Fig. 4. X-ray diffraction patterns of films of PEO and the SPEs with bentonite-Li⁺ and with the nanocomposite PEO@bentonite-Li⁺. PEO Mw 600,000.

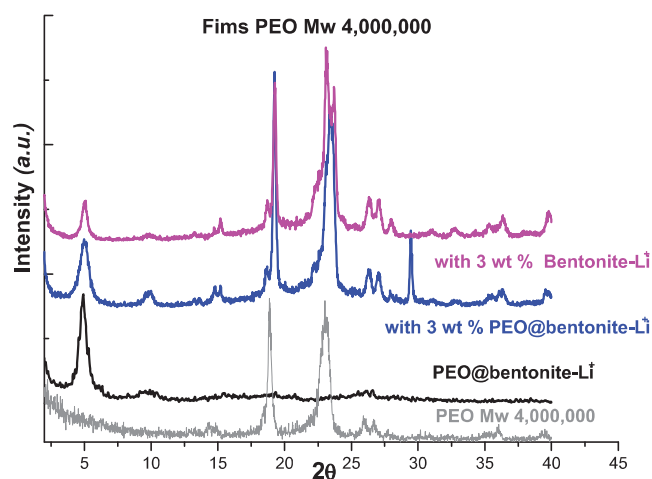


Fig. 5. X-ray diffraction patterns of films of PEO and the SPEs with bentonite-Li⁺ and with the nanocomposite PEO@bentonite-Li⁺ with PEO Mw 4,000,000.

intercalation of PEO into the clay occurs spontaneously. So the filler in this composite leads also to a PEO/clay nanocomposite similar to that previously prepared with a polymer solution described above. This agrees with the preparation of hybrid clay nanocomposites in organic melts widely studied by Gianelis et al. [27]. Certainly, there is large affinity of PEO for alkali cations confined in this kind of layered polyanionic aluminosilicates. In the diffraction patterns of the SPEs prepared using the pre-intercalated clay as filler, basal interlayer spacing of PEO@bentonite-Li⁺ nanocomposite is observed at $2\theta = 4.88$. Thus, the insertion of nanocomposite in the PEO matrix causes a slight decrease of ca. 0.5 Å, in its interlaminar distance (18.0 Å), probably due to a slight loss of PEO in the interlaminar space. A different behavior was observed in previously reported SPEs containing organic modified clay as filler, with clay concentration (2.9%) similar to ours. In that case, a notorious increment of the interlayer distance, even undergoing swelling with totally dispersed clay sheets was observed [28]. We believe this does not occur in our case because of both, the strength of the interaction clay/polymer and the commensurate nature of PEO/bentonite nanocomposites which prevents further insertion of polymer.

Similarly, the diffraction patterns of PEO and SPE membranes using PEO with molecular weight 4,000,000 are compared in Fig. 5. Results are in general similar to those with PEO 600,000. However observed interlaminar distances for PEO/clay nanocomposites, namely 17.3 and 17.7 Å for the addition of bentonite-Li⁺ and PEO@bentonite-Li⁺ as filler respectively, indicate that for the polymer with higher molecular weight *in situ* intercalation is also effective. Concerning the polymer, the characteristic diffraction peaks of crystalline PEO, observed between $2\theta = 15^\circ$ and 30° [29] become broader in the composite polymer electrolyte. This agrees with results reported for the addition of comparable amount of filler (ca. 3%) [30].

3.2.2. Optical properties

All PEO or SPE materials prepared led to homogenous, flexible and transparent films. Although transparency of the materials is in general less important for energy storage, there are other applications, for instance, photoelectrochemical processes in which the transparency of electrolyte and other components of the cell are important. Fig. 6 shows relative transmittances of films containing bentonite-Li⁺ and the nanocomposite PEO@bentonite-Li⁺ as filler, compared with that of a film containing only PEO. Interestingly, the transparency of the films increases with the addition of the filler. This effect is especially high when the SPE is prepared with PEO@bentonite-Li⁺ as filler. In general, transparency in

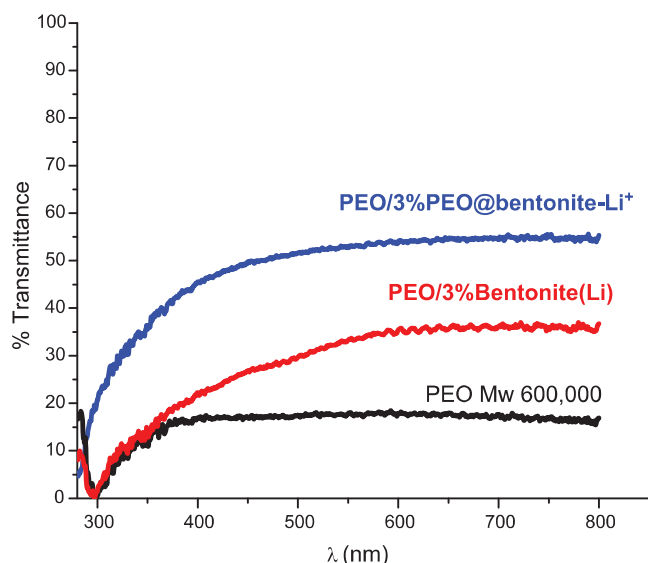


Fig. 6. UV-visible spectra of films of the polymer and SPEs with bentonite- Li^+ and PEO@bentonite- Li^+ with PEO Mw 600,000.

complex polymer systems may be considered a consequence of the molecular compatibility between the components, as well as of the degree of polymer crystallinity. In the case of composites of polymers with inorganic additives, variables like filler grain size, affinity between organic and inorganic components, and homogeneity of the suspension of the filler in the polymer also have to be considered.

3.2.3. Mechanical properties

In general, solid polymer electrolytes using only PEO have turned out to be inadequate for practical application in lithium-ion batteries due to their low homogeneity, bad mechanical properties and poor conductivity [31,32]. In this work, the effect of the addition of the fillers bentonite- Li^+ and the nanocomposite PEO@bentonite- Li^+ on the mechanical properties of resulting SPEs was tested. As summarized in Table 2, Young's modulus (70 MPa), tensile strength (0.7 MPa), and the toughness (0.75 MPa), observed in the film containing only PEO Mw 600,000, are improved significantly with the addition of the fillers. With the incorporation of 3 wt% bentonite- Li^+ these parameters increase, reaching values which in the case of tensile strength and fail strain are about 8 times higher than those of PEO. Interestingly, the film prepared using a dispersion of 3 wt% PEO@bentonite- Li^+ results in further improvement beyond those with the unmodified clay. Thus, toughness results to be 20 and 5 times better than those of pure PEO and bentonite- Li^+ respectively. Fig. 7 shows the tensile stress-strain curves for PEO Mw 600,000 SPE films.

Experiments performed using PEO Mw 4,000,000 show that in this system the effect of the filler is not too significant. Values are in general similar for all the films and, in some cases, a slight deterioration of mechanical properties with the addition of fillers is observed.

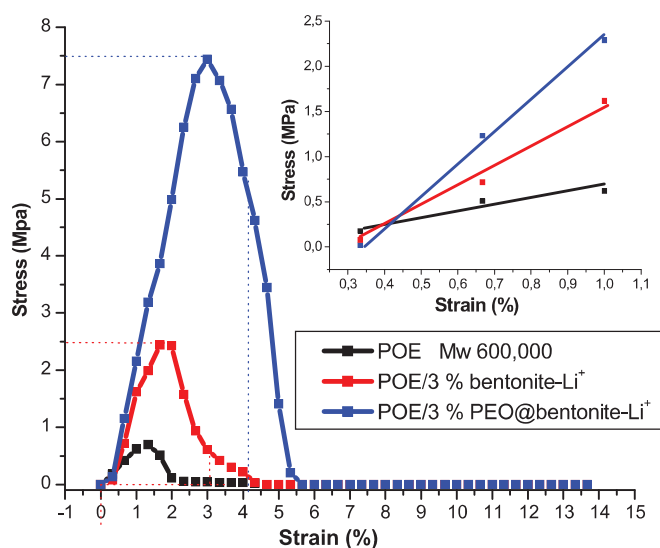


Fig. 7. Tensile stress-strain behavior of films of PEO and SPEs with bentonite- Li^+ and PEO@bentonite- Li^+ with PEO Mw 600,000.

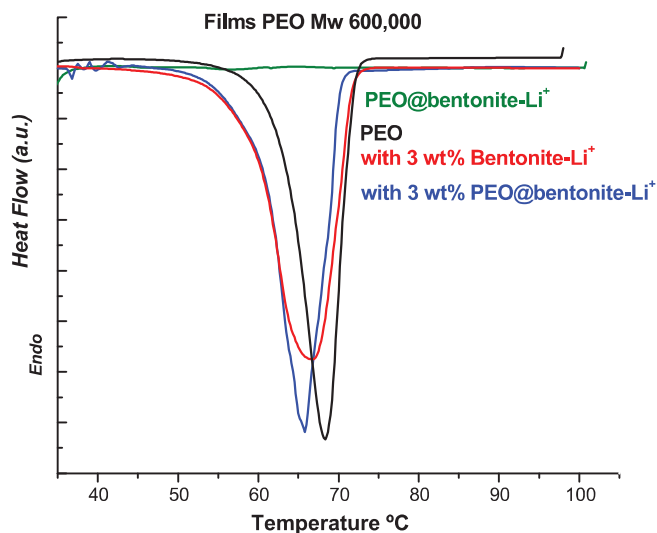


Fig. 8. Differential scanning calorimetry measurement of PEO and SPEs with bentonite- Li^+ and PEO@bentonite- Li^+ with PEO Mw 600,000.

3.2.4. Differential scanning calorimetry

Before further discussing the effect of the nature of fillers used in this work on the properties of the resulting SPEs, it is interesting to analyze the crystallinity of PEO in the various samples as detected by differential scanning calorimetry (DSC) measurements displayed in Fig. 8. These thermograms and the corresponding ΔH values consigned in Table 3, show slight changes in crystallinity, near or even lower than 10% [33]. The variations observed do not correlate with the values of conductivity, especially in the case of PEO with lower molecular weight, where the addition of

Table 2

Mechanical properties of PEO films containing 3 wt% PEO@bentonite- Li^+ and bentonite- Li^+ . PEO Mw 600,000 and 4,000,000.

Sample	Young's modulus (MPa)	Tensile strength (MPa)	Fail strain (%)	Toughness (MPa)
PEO Mw 600,000	70 ± 7	0.7 ± 0.07	1.7 ± 0.17	0.75 ± 0.075
PEO 600,000/3 wt% bentonite- Li^+	247 ± 24.7	2.9 ± 0.29	3.4 ± 0.3	6.3 ± 0.63
PEO 600,000/3 wt% PEO@bentonite- Li^+	325 ± 32.5	7.5 ± 0.75	4.2 ± 0.42	21.3 ± 2.1
PEO Mw 4,000,000	231 ± 23.1	9.4 ± 0.94	6.2 ± 0.62	37.5 ± 3.75
PEO 4,000,000/3 wt% bentonite- Li^+	285 ± 28.5	7.5 ± 0.75	4.1 ± 0.41	18.2 ± 1.82
PEO 4,000,000/3 wt% PEO@bentonite- Li^+	252 ± 25.2	8.5 ± 0.85	4.6 ± 0.46	24.5 ± 2.45

Table 3

Thermodynamic properties and crystallinity of PEO Mw 600,000 and Mw 4,000,000 films containing 3 wt% PEO@bentonite-Li⁺ and bentonite-Li⁺.

Sample	T _f (°C)	ΔH (J/g)	Crystallinity ^a
PEO Mw 600,000	68.29	127.2	59.5
PEO 600,000/3 wt% bentonite-Li ⁺	66.30	147.2	68.9
PEO 600,000/3 wt% PEO@bentonite-Li ⁺	65.41	141.2	66.0
PEO Mw 4,000,000	68.78	117.8	54.3
PEO 4,000,000/3 wt% bentonite-Li ⁺	69.19	138.9	65.0
PEO 4,000,000/3 wt% PEO@bentonite-Li ⁺	65.15	105.4	49.3

^a %X = (ΔH_{sample}/213.7) × 100 [33].

bentonite-Li⁺ always appears to enhance polymer crystallinity. This increment is however moderated by previous intercalation of PEO into the clay, this being particularly effective in PEO Mw 4,000,000 where the addition of PEO@bentonite-Li⁺ leads to a net thought slight decrease of crystallinity respect to that of neat PEO.

3.2.5. Electrical conductivity

Electrical conductivity values measured at room temperature for PEO and both SPEs, with bentonite-Li⁺ and bentonite-Li⁺ intercalated with the same polymer, are reported in Table 4. Conductivities of neat PEO with molecular weights 600,000 and 4,000,000 are 1.2×10^{-9} and 2.5×10^{-9} S cm⁻¹ respectively. Observed conductivities are in general rather poor. However, they are similar to some reported in the literature [34] and correspond to those expected for SPE samples with low content of lithium (EO:Li⁺ < 34:1), inserted by diffusion from its salt at room temperature, and without any further thermal annealing. PEO-based SPEs with higher conductivity values may be certainly attained by using higher EO:Li⁺ ratios and appropriate thermal treatments [10]. Nevertheless SPE samples for this study were prepared under conditions specially selected for avoiding the formation of crystalline PEO/lithium-ion complexes. So, the effect of the filler will be mainly due to changes in the polymer phases and eventual synergic effects caused by intercalation of the polymer in the layered filler might be better evaluated.

The addition of bentonite-Li⁺ produces, in both polymers, an increment in the conductivity of about one order of magnitude. The comparison of these results with previously reported data is difficult because the effect of ceramic additives on the conductivity, as in the ceramic free polymer electrolytes, is also influenced by both lithium concentration and thermal history of the sample. Indeed, Fullerton-Shirey and Maranas [13] have recently demonstrated that in the system PEO/CIO₄/Al₂O₃ the effect of nanoparticles on the conductivity is maximized at a salt concentration nearly the eutectic concentration of the unfilled SPE; i.e. at a EO:Li⁺ ratio of 10:1, where the addition of 5% Al₂O₃ improves the conductivity in about one order of magnitude at 30 °C. Although ceramic-filled, PEO-based polymer electrolytes are normally studied at lithium contents yielding maximal conductivity, far from the concentration used in this study, our results are consistent with reported conductivity improvements using as additives conventional ceramic nanoparticles [35], clays [14] and organoclays [16,36], which in general are between one and two orders of magnitude.

Table 4

Conductivities of PEO Mw 600,000 and Mw 4,000,000 SPE films.

Sample	σ (S cm ⁻¹) at 25 °C
PEO Mw 600,000	1.23×10^{-9}
PEO 600,000/3 wt% bentonite-Li ⁺	3.89×10^{-8}
PEO 600,000/3 wt% PEO@bentonite-Li ⁺	1.81×10^{-7}
PEO Mw 4,000,000	2.45×10^{-9}
PEO 4,000,000/3 wt% bentonite-Li ⁺	1.79×10^{-8}
PEO 4,000,000/3 wt% PEO@bentonite-Li ⁺	2.48×10^{-8}

The effect of the presence of PEO@bentonite-Li⁺ as filler is notorious in the case of the SPE with PEO 600,000, where a conductivity of 1.81×10^{-7} S cm⁻¹ is attained, i.e. an increase of about two orders of magnitude respect to neat PEO and more than 5 times that of bentonite-Li⁺. On the other hand, in the SPE with PEO Mw 4,000,000 the effect of both fillers, bentonite-Li⁺ and PEO@bentonite-Li⁺, is practically the same. To the best of our knowledge this is the first time that such an effect is so clearly observed in this kind of solid polymer electrolytes.

There are a number of factors involved in the effect of ceramic fillers on the conductivity of polymer electrolytes. Most of them are expected to arise from the electrostatic and Lewis acid–base interactions occurring between functional groups located on the surface of the ceramic particles and the polymer electrolyte components [37]. The main ways by which these interactions affect the conductivity appear to be changes of crystallinity and segmental movements of the polymer, and changes in the activity of the ionic carrier. However in our particular case, there is an interesting situation which may be useful for understanding interaction equilibria in such rather complex systems. Indeed we are observing significant differences in both conductivity and crystallinity between two systems which, due to spontaneous intercalation of the bentonite-Li⁺ during SPE preparation, happen to have practically the same filler species, i.e. PEO@bentonite-Li⁺ in both cases.

The addition of layered active fillers, like layered clays containing lithium, may contribute to the activity of the ionic carrier essentially by increasing the local concentration of electroactive species as well as by competing with the polymer and salt counter anion for the coordination of lithium cations [38]. Considering the low donor ability of PF₆⁻ ion and the fact that in the filler PEO@bentonite-Li⁺ the lithium-ion environment is not very different from that in the PEO matrix, the latter of these factors should be less important in our case. Therefore the conductivity enhancement caused by filler assisted carrier activity should be approximately the same in both SPEs analyzed here. Thus, differences in observed conductivities between these SPEs have to arise from changes in crystallinity which, as evidenced by DSC measurements discussed above, is different in both cases. Such a different activity of both fillers is caused not by the nature of the filler but by the procedure through which they have been produced. We believe that the main difference between the two fillers used in our experiments is their compatibility with the polymer matrix. Compared with unmodified bentonite, PEO@bentonite-Li⁺ is certainly more compatible with PEO, so in a PEO suspension this filler will end up being more highly and homogeneously dispersed, particularly in the polymer with lower molecular weight [12]. Because of its higher affinity with PEO, this nanocomposite could reach more easily crystalline domains of the polymer, thus altering its structure and producing more favorable conditions for ionic conductivity. Unmodified bentonite, in turn, would be mostly located in amorphous microphases of the polymer where, as deduced from data in Table 2, the addition of this filler produces an increment of crystallinity, probably due to micro dynamic ordering processes produced during the intercalation of the clay in the melt.

Thus, differences in the degree of micro homogeneity appear to be an important factor in determining the behavior in PEO/clay based systems, offering a coherent explanation for the effect of tested layered inorganic additives on the properties of this kind of SPE membranes. This explanation allows us to understand the observed electrical, thermal, optical, and mechanical properties of the SPEs obtained.

In conclusion, results described here show that by considering thermodynamic factors including interactions among SPE components, together with kinetic effects associated to preparation procedures and possible variations of resulting micro homogeneity in composite polymer electrolytes, it is possible to understand

many aspects of the conductivity, structure, and properties in PEO/layered clay systems.

4. Conclusions

The preparation of solvent free composite polymer electrolytes based on two different poly(ethylene oxide), with molecular weight 600,000 and 4,000,000, using as ceramic additive lithiated bentonite and the same clay, but intercalated with the polymer used as matrix (PEO@bentonite-Li⁺), leads to self-supported transparent films. In the case of PEO of lower molecular weight, the presence of the filler induces significant improvements on the mechanical properties and conductivity of the films.

Independently of the pristine filler, the nature of the clay particles detected results to be the same in both SPEs, i.e. bentonite intercalated with the polymer. This is due to the spontaneous *in situ* intercalation of bentonite while preparing the films. The mechanical properties, transparency, and conductivity of the films obtained using the nanocomposite are better than those obtained when using the unmodified clay as filler.

Changes in the polymer matrix and consequently the effects on the properties of the SPE attained by addition of ceramic filler may be explained considering electrostatic and Lewis acid–base interactions between functional groups located on the surface of the ceramic particles and the polymer electrolyte components excluding, due to used experimental conditions, any important participation of crystalline PEO–Li⁺ complexes. However, differences in the behavior of the SPEs, arising from the way the clay nanocomposite is formed, may be understood by principally considering the higher affinity between the pre-modified clay and PEO matrix. Thus, a higher micro homogeneity may be reached by using PEO@bentonite-Li⁺ as filler. Since the higher chemical affinity of this filler with PEO allows for an easier reach and modification of crystalline PEO domains, whereas the action of unmodified clay would be mainly limited to the amorphous domains of the polymer, thus increasing the micro heterogeneity of the system. Since such processes are essentially kinetic, they are observed only in matrices of polymers of relatively low molecular weight.

The present work has shown how controlling the homogeneity in this type of polymer electrolytes may be critical for the fabrication of improved solid electrolytes. However further knowledge is still needed, for instance, on the effect of the concentration of filler. Further research in that direction is still in progress.

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